Dhruv Shah

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA, 2025 – 2029

B.S. in Electrical and Computer Engineering, minor in AI — QPA: 3.73

Relevant Coursework: Digital Systems, Principles of Imperative Computation (Data Structures & Algorithms), Concepts of Mathematics, Linear Algebra, Probability Theory

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TECHNICAL EXPERIENCE

AI/ML Medical Imaging Intern | Netramind Innovations — May 2026 – August 2026

- Developing U-Net-based deep learning models for retinal image segmentation using Python and PyTorch, targeting automated ischemia detection in fluorescein angiography images.
- Designing preprocessing, augmentation, training, and evaluation pipelines for medical image segmentation datasets and benchmarking model performance using Dice and IoU metrics.
- Conducted experiments across model architectures, image resolutions, and loss functions to improve segmentation accuracy on challenging clinical imaging datasets, working towards authoring a research paper.

Software Engineering Intern | Lockheed Martin — May 2025 – July 2025

- Created a Python service and REST APIs for Stable Diffusion XL (SDXL) image generation, supporting configurable inference parameters, image serialization, and workflow automation.
- Built a LangChain-style abstraction for integrating large language models and diffusion models, enabling agentic workflows and model orchestration.
- Implemented image preprocessing and postprocessing pipelines using Pillow, BytesIO, and JSON-based API interfaces, improving robustness and customizability of generated images.

Computational Research Assistant | University of Florida — May 2024 – July 2024

- Developed numerical simulation software in Python and FORTRAN to model transonic airfoil behavior and analyze aerodynamic performance under varying flight conditions.

PROJECTS

Retinal Ischemia Segmentation — 2026

- Trained and evaluated U-Net++, MANet, and DeepLabV3-based segmentation models.
- Implemented preprocessing and experimentation pipelines including CLAHE, top-hat filtering, Gaussian enhancement, custom loss functions, and Dice/IoU-based model evaluation.

iOS Application — 2023

- Designed and deployed a Swift-based iOS application on the App Store w/ 200+ downloads across 4 continents.

SKILLS

Languages: Python, Java, C/C++, JavaScript, Swift

AI/ML: PyTorch, TensorFlow/Keras, OpenCV, Deep Learning, Image Segmentation, U-Net++, MANet, Diffusion Models (SDXL), Large Language Models (LLMs), LangChain, LangGraph, Agentic AI Workflows, model evaluation

Software Engineering: Object-Oriented Programming, REST APIs, Git/GitHub, Linux, Agile/JIRA, Data Structures & Algorithms, NumPy, Pandas, Jupyter, Pillow